Below is a **day-to-day (real-world) "most used" list of Cloud SDK classes + methods** for **AWS, Azure, and GCP**, with **real application use-cases** and **mini examples** (Java + Python style), in **table format**.

✅ Notes

- "All classes/methods of all versions" is impossible (cloud SDKs are huge + constantly changing).
- So I'm listing **the most commonly used SDK client classes + key methods** in **production apps** (microservices, ETL, backend APIs, analytics, DevOps automation).
- AWS examples are from **AWS SDK Java v2** and **Boto3**, Azure from **azure-sdk-for-java**, GCP from **google-cloud-java**. (AWS Documentation)

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1) COMMON CLOUD SDK PATTERN (Applies to AWS/Azure/GCP)

What you do	AWS SDK	Azure SDK	GCP SDK	Real use case
Create client	S3Client.builder()	new BlobServiceClientBuilder()	StorageOptions.getDefaultInstance().getService()	Create service client in Spring Boot / Django
Authenticate	IAM role / Access Keys	TokenCredential (Managed Identity / SPN)	Service account / ADC	Run securely in VM / K8s
Request object	PutObjectRequest.builder()	BlobClient.upload() request	BlobInfo.newBuilder()	Controlled payload + metadata
Handle errors	SdkException	HttpResponseException	ApiException	Retry / fallback

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2) AWS (Java SDK v2 + Boto3) — Most used classes + methods

AWS SDK Java v2 uses service clients like S3Client, DynamoDbClient, SqsClient, etc. (AWS Documentation)
Boto3 provides Python APIs and clients/resources. (Boto3)

AWS — Storage (S3)

Service	Common Class	Common Methods used daily	Real-time Use Cases	Example
Amazon S3	Java: S3Client	putObject(), getObject(), listObjectsV2(), deleteObject(), headObject()	Upload user files, profile pics, invoices, reports	**Java**: s3.putObject(PutObjectRequest..., RequestBody.fromFile(...)) (AWS Documentation)
Amazon S3	Python: boto3.client("s3")	upload_file(), download_file(), list_objects_v2(), head_object()	Media storage, backups, static website hosting	**Python**: s3.upload_file("a.jpg","bucket","a.jpg") (AWS Documentation)

AWS — Database (DynamoDB)

Service	Common Class	Common Methods	Real-time Use Cases	Example
DynamoDB	Java: DynamoDbClient	putItem(), getItem(), query(), scan(), updateItem(), deleteItem()	User session store, product catalog, IoT events	ddb.getItem(GetItemRequest.builder()...) (AWS Documentation)
DynamoDB	Python: boto3.resource("dynamodb").Table()	put_item(), get_item(), query(), update_item()	Same as above	table.query(KeyConditionExpression=...)

AWS — Messaging (SQS / SNS)

Service	Class	Common Methods	Real use case
SQS	Java: SqsClient	sendMessage(), receiveMessage(), deleteMessage()	Async processing: order processing, email sender, invoice

			generator
SNS	Java: SnsClient	publish(), subscribe(), createTopic()	Push notifications, fanout to multiple services

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AWS — Serverless (Lambda)

Service	Class	Common Methods	Real-time Use Cases	Example
Lambda	Java: LambdaClient	invoke(), getFunction(), updateFunctionCode()	Trigger ML inference, async workflows	lambda.invoke(InvokeRequest.builder()...) (AWS Documentation)

AWS — Secrets & Config

Service	Class	Common Methods	Use cases
Secrets Manager	SecretsManagerClient	getSecretValue(), createSecret(), putSecretValue()	Store DB password, API keys
Parameter Store (SSM)	SsmClient	getParameter(), putParameter(), getParametersByPath()	Central config for microservices

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AWS — Identity & Auth (IAM + STS)

Service	Class	Common Methods	Use cases
STS	StsClient	assumeRole(), getCallerIdentity()	Cross-account access, CI/CD deployments
IAM	IamClient	createRole(), attachRolePolicy()	Automation & provisioning

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AWS — Compute / Containers

Service	Class	Common Methods	Use cases
EC2	Ec2Client	runInstances(), describeInstances(), terminateInstances()	Infra automation
ECR	EcrClient	getAuthorizationToken(), describeRepositories()	Docker image registry
ECS / EKS	EcsClient / Kubernetes APIs	runTask(), describeServices()	Deploy services

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3) AZURE (Java SDK) — Most used classes + methods

Azure apps most commonly use:

- **Storage SDK** (Blob)
- **Identity** (TokenCredential)
- **Cosmos DB**
- **Resource Manager (ARM)** for provisioning

Azure Blob SDK uses BlobServiceClient created via BlobServiceClientBuilder. (Microsoft Learn)

Azure — Storage (Blob)

Service	Common Class	Common Methods	Real-time Use Cases	Example
Azure Blob Storage	BlobServiceClient	getBlobContainerClient()	Access containers for uploads/downloads	BlobContainerClient c = svc.getBlobContainerClient("docs"); (Microsoft Learn)
Azure Blob Storage	BlobContainerClient	getBlobClient(), listBlobs()	List all files / cleanup jobs	container.listBlobs()
Azure Blob Storage	BlobClient	upload(), download(), delete()	Upload invoices/resumes, download reports	blob.upload(fileInputStream, length, true) (Microsoft Learn)

Azure — Identity (Authentication)

Library	Class	Common Methods	Use cases
azure-identity	DefaultAzureCredential	token auto-resolution	Runs on local + VM + AKS with same code
azure-core	TokenCredential	passed into service client builder	Secure authentication in enterprise

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Azure — Database (Cosmos DB)

Service	Class	Common Methods	Use cases
Cosmos DB	CosmosClient / CosmosContainer	createItem(), readItem(), queryItems(), replaceItem()	NoSQL apps: chat app, profile store

(For management/provisioning Cosmos resources you also use ARM libraries.) (Microsoft Learn)

Azure — Messaging (Service Bus)

Service	Class	Common Methods	Use cases
Service Bus	ServiceBusClientBuilder	sender(), receiver()	Queue-based async processing (like SQS)

Azure — Resource provisioning (ARM)

Area	Class	Common Methods	Use cases
Azure Resource Manager	AzureResourceManager	authenticate(), .subscriptions(), .resourceGroups()	DevOps automation / Infra-as-code style provisioning (GitHub)

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4) GCP (Google Cloud Java) — Most used classes + methods

Google recommends **Cloud Client Libraries** for Java. (Google Cloud Documentation)
BigQuery client library provides APIs for datasets/tables/jobs. (Google Cloud Documentation)
Pub/Sub is used for event streaming and integrations. (Google Cloud)

GCP — Storage (Cloud Storage)

Service	Common Class	Common Methods	Use cases	Example
Cloud Storage	com.google.cloud.storage.Storage	create(), get(), list(), delete()	Upload files, backups, CDN static content	storage.create(BlobInfo, byte[])

GCP — BigQuery (Data warehouse / analytics)

Service	Common Class	Common Methods	Real use case
BigQuery	BigQuery	query(), create(), getTable(), insertAll()	Analytics dashboards, reporting systems, ETL pipelines (Google Cloud Documentation)

GCP — Messaging (Pub/Sub)

Service	Common Class	Common Methods	Use cases
Pub/Sub	Publisher, Subscriber	publish(), message receiver callbacks	Event-driven microservices, stream pipelines to BigQuery/Dataflow (Google Cloud)

GCP — Firestore (NoSQL)

Service	Common Class	Common Methods	Real use case
Firestore	Firestore, DocumentReference	collection(), document(), set(), get(), update()	Realtime chat app, user profile store (Firebase)

GCP — Compute / Serverless

Service	Class	Common Methods	Use cases
Cloud Functions	HTTP-triggered function	handler methods	lightweight serverless
Cloud Run	container deploy	revision traffic	deploy microservices without K8s

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5) REAL-TIME CLOUD APP EXAMPLES (Which services used where)

| Application Type | AWS | Azure | GCP ||---|---|---|
| File upload backend (docs/images) | S3 + CloudFront | Blob Storage + CDN | Cloud Storage + Cloud CDN |

| Order processing microservices | SQS + Lambda + DynamoDB | Service Bus + Functions + Cosmos DB | Pub/Sub + Cloud Run + Firestore |

| Analytics pipeline | Kinesis/S3 + Athena/Redshift | Event Hub + Synapse | Pub/Sub + Dataflow + BigQuery (Google Cloud) |

| Secrets/config for services | Secrets Manager / SSM | Key Vault / App Config | Secret Manager |

If you want, I can also provide:
✅ **"AWS vs Azure vs GCP SDK cheat sheet"** (Spring Boot + Java),
✅ **Best practices** (retry/backoff, pagination, batching),
✅ A **full runnable project** with *AWS + Azure + GCP clients in one Java app* (like a cloud abstraction layer).